

SECTION 11

ORDERING INFORMATION AND MECHANICAL DATA

This section contains the pin assignments and package dimensions of the MC68020 and the MC68EC020. In addition, detailed information is provided to be used as a guide when ordering.

11.1 STANDARD ORDERING INFORMATION

11.1.1 Standard MC68020 Ordering Information

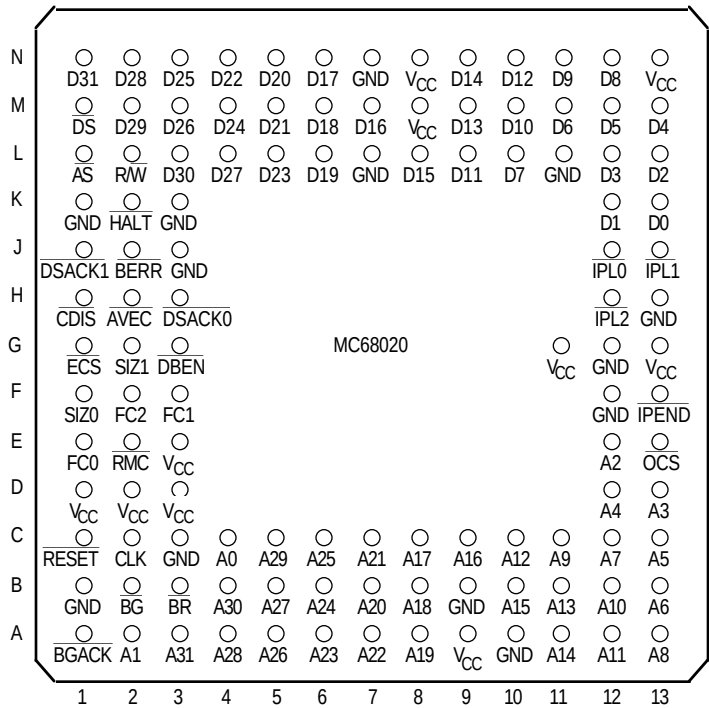
Package Type	Frequency (MHz)	Temperature (°C)	Order Number
Ceramic Pin Grid Array RC Suffix	16.67	0 to 70	MC68020RC16
	20.0	0 to 70	MC68020RC20
	25.0	0 to 70	MC68020RC25
	33.33	0 to 70	MC68020RC33
Plastic Quad Flat Pack FC Suffix	16.67	0 to 70	MC68020FC16
	20.0	0 to 70	MC68020FC20
	25.0	0 to 70	MC68020FC25
Plastic Pin Grid Array RP Suffix	16.67	0 to 70	MC68020RP16
	20.0	0 to 70	MC68020RP20
	25.0	0 to 70	MC68020RP25
Ceramic Quad Flat Pack FE Suffix	16.67	0 to 70	MC68020FE16
	20.0	0 to 70	MC68020FE20
	25.0	0 to 70	MC68020FE25
	33.33	0 to 70	MC68020FE33

11.1.2 Standard MC68EC020 Ordering Information

Package Type	Frequency (MHz)	Temperature (°C)	Order Number
Plastic Pin Grid Array RP Suffix	16.67	0 to 70	MC68EC020RP16
	25.0	0 to 70	MC68EC020RP25
Plastic Quad Flat Pack FG Suffix	16.67	0 to 70	MC68EC020FG16
	25.0	0 to 70	MC68EC020FG25

11.2 PIN ASSIGNMENTS AND PACKAGE DIMENSIONS

11.2.1 MC68020 RC and RP Suffix—Pin Assignment

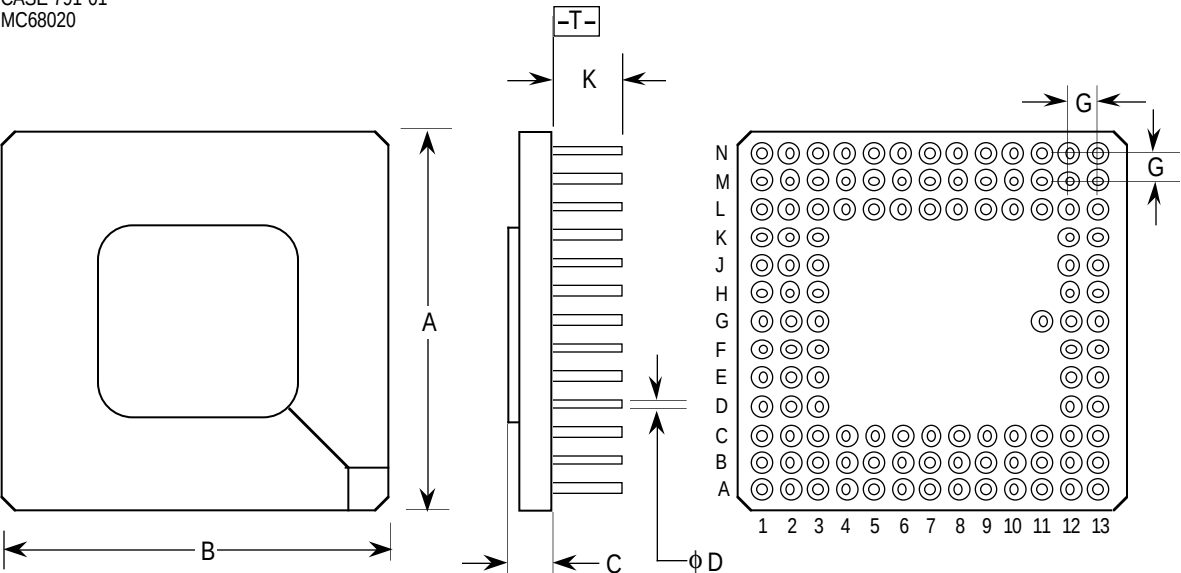


The V_{CC} and GND pins are separated into four groups to provide individual power supply connections for the address bus buffers, data bus buffers, and all other output buffers and internal logic. It is recommended that all pins be connected to power and ground as indicated.

Group	V _{CC}	GND
Address Bus	A9, D3	A10, B9, C3, F12
Data Bus	M8, N8, N13	L7, L11, N7, K3
Logic	D1, D2, E3, G11, G13	G12, H13, J3, K1
Clock	—	B1

11.2.2 MC68020 RC Suffix—Package Dimensions

RC SUFFIX
CASE 791-01
MC68020



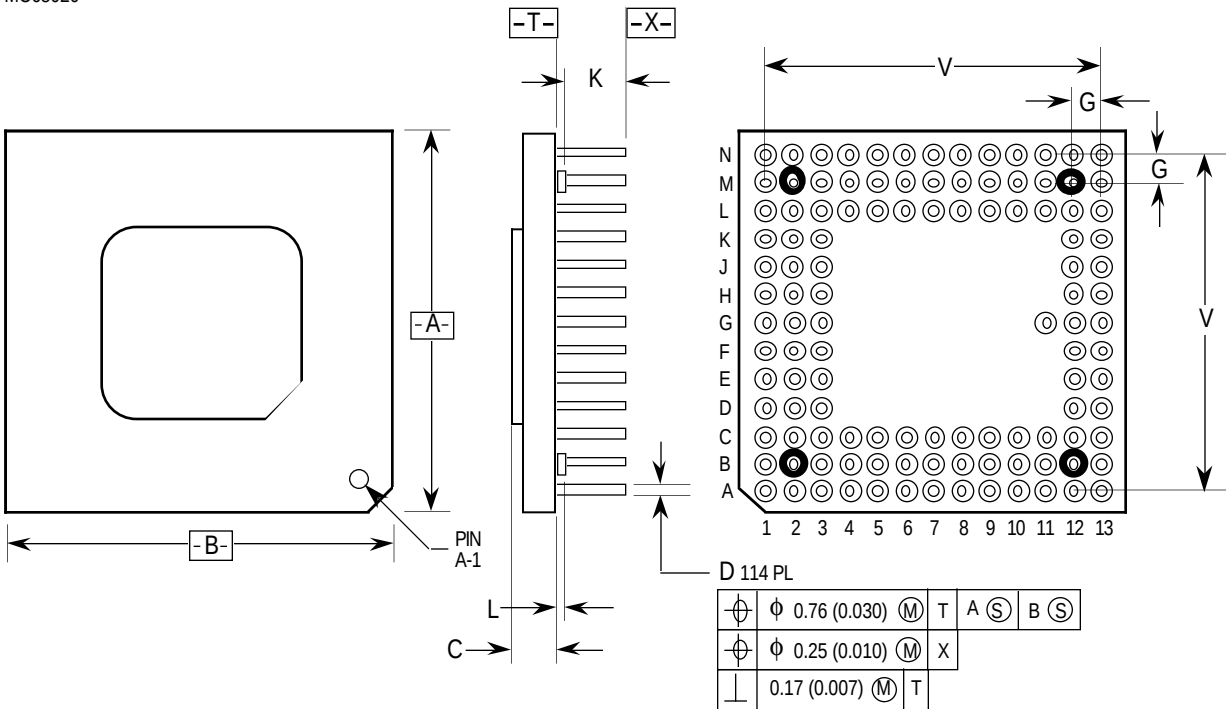
DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	34.04	35.05	1.340	1.380
B	34.04	35.05	1.340	1.380
C	2.54	3.81	0.100	0.150
D	0.43	0.55	0.017	0.022
G	2.54 BSC		0.100 BSC	
K	4.32	4.95	0.170	0.195

- NOTES:
1. A AND B ARE DATUMS AND T IS A DATUM SURFACE.
 2. POSITIONAL TOLERANCE FOR LEADS (114 PLACES).
 3. DIMENSIONING AND TOLERANCING PER Y14.5M,1982.
 4. CONTROLLING DIMENSION: INCH.

$\varnothing$	0.13 (0.005)	(M)	T	A	(S)	B	(S)
---------------	--------------	-----	---	---	-----	---	-----

11.2.3 MC68020 RP Suffix—Package Dimensions

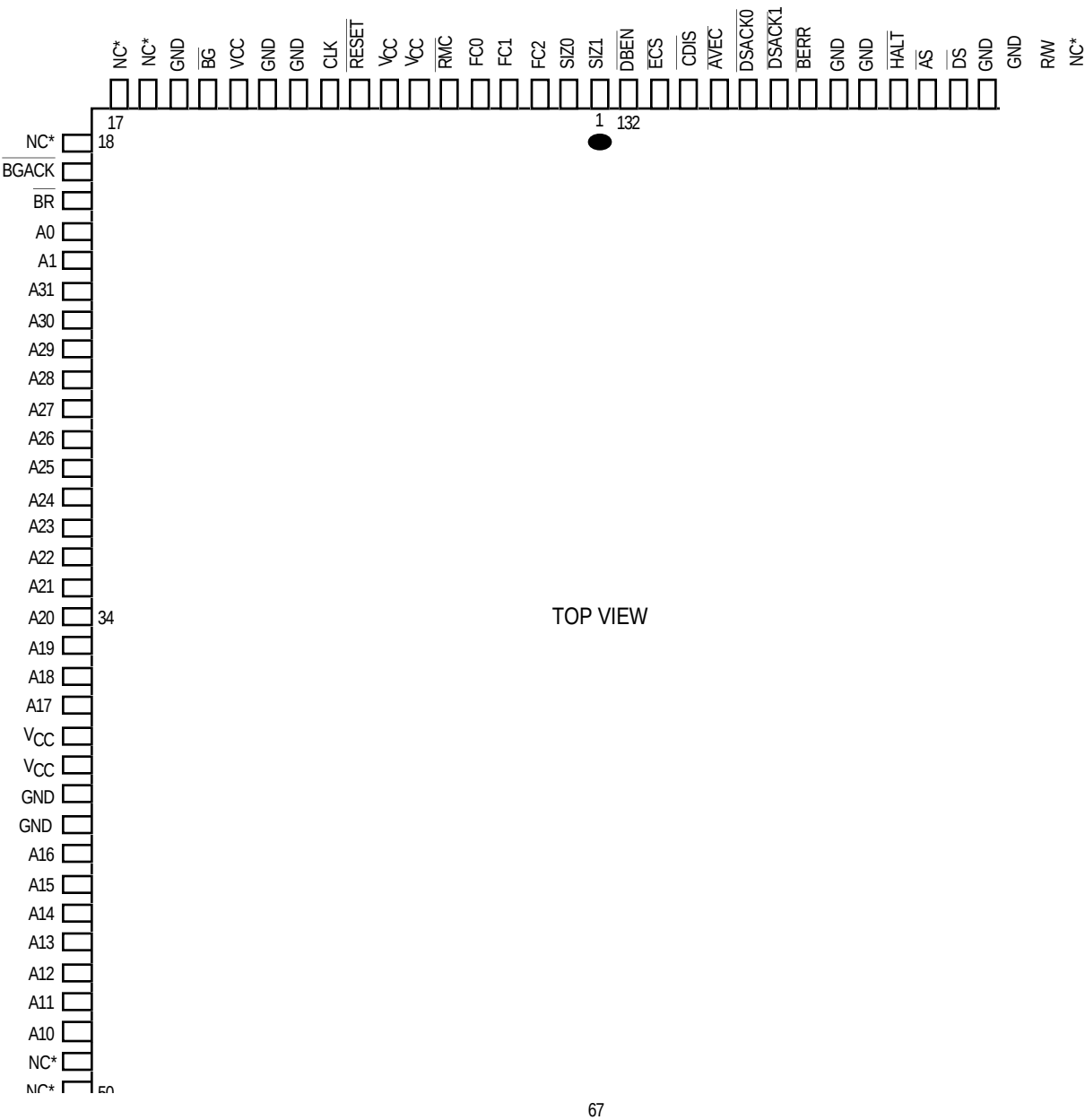
RP SUFFIX
CASE 789E-02
MC68020



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	34.04	35.05	1.340	1.380
B	34.04	35.05	1.340	1.380
C	2.92	3.18	0.115	0.135
D	0.44	0.55	0.017	0.022
G	2.54 BSC		0.100 BSC	
K	2.79	3.81	0.110	0.150
L	1.02	1.52	0.040	0.060
V	30.48 BSC		1.200 BSC	

- NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
 2. CONTROLLING DIMENSION: INCH.
 3. DIMENSION D INCLUDES LEAD FINISH.
 4. 789E-01 OBSOLETE. NEW STANDARD 789E-02.

11.2.4 MC68020 FC and FE Suffix—Pin Assignment

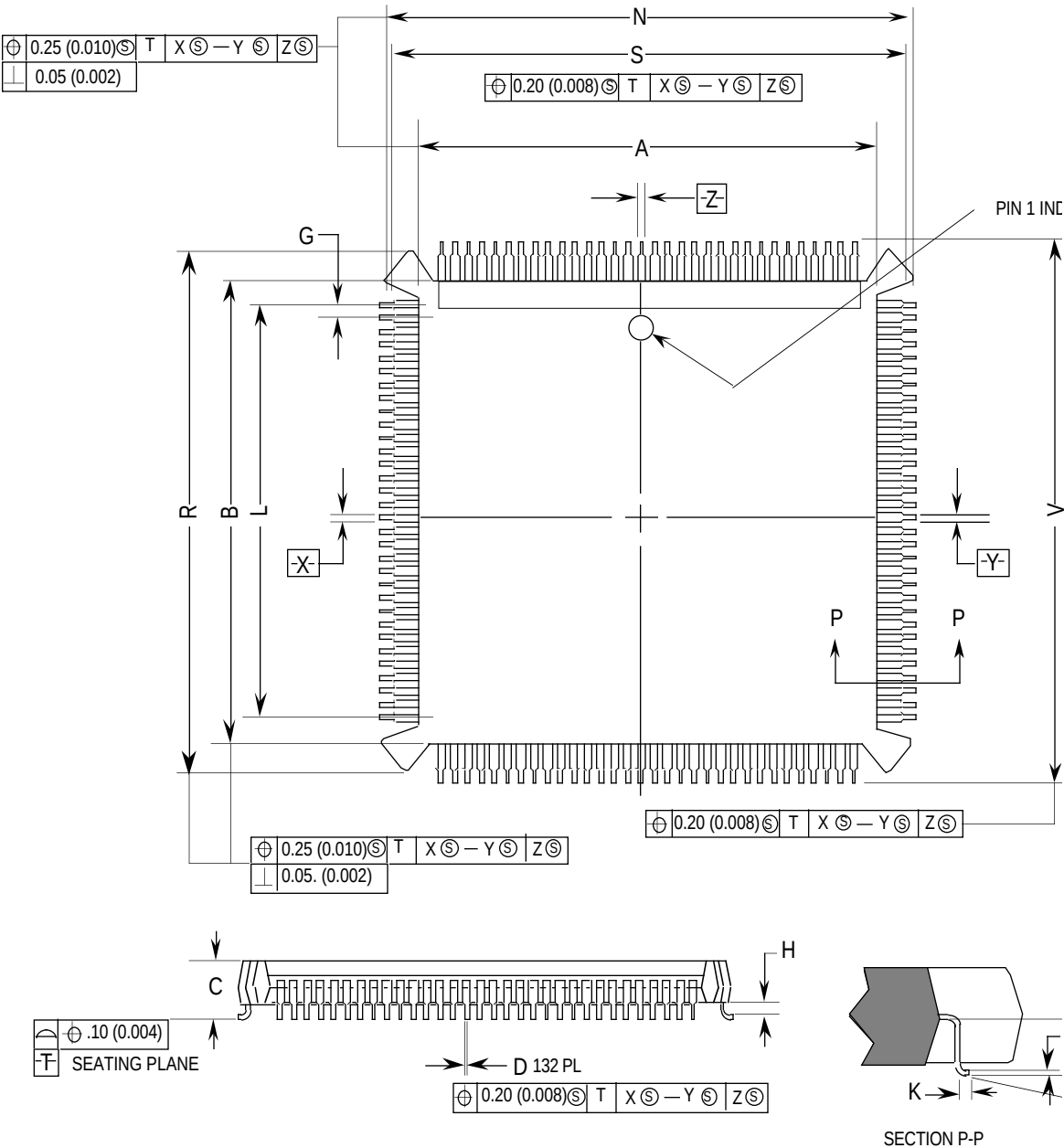


The V_{CC} and GND pins are separated into four groups to provide individual power supply connections for the address bus buffers, data bus buffers, and all other output buffers and internal logic. It is recommended that all pins be connected to power and ground as indicated. NC pins are reserved by Motorola for future use and should have no external connection.

Group	V _{CC}	GND
Address Bus	13, 38, 39	15, 40, 41, 62
Data Bus	79, 80, 96, 97	77, 78, 98, 99, 119, 120
Logic	7, 8, 65, 66	67, 68, 124, 125
Clock	—	11, 12

11.2.5 MC68020 FC Suffix—Package Dimensions

FC SUFFIX
CASE 831A-01
MC68020

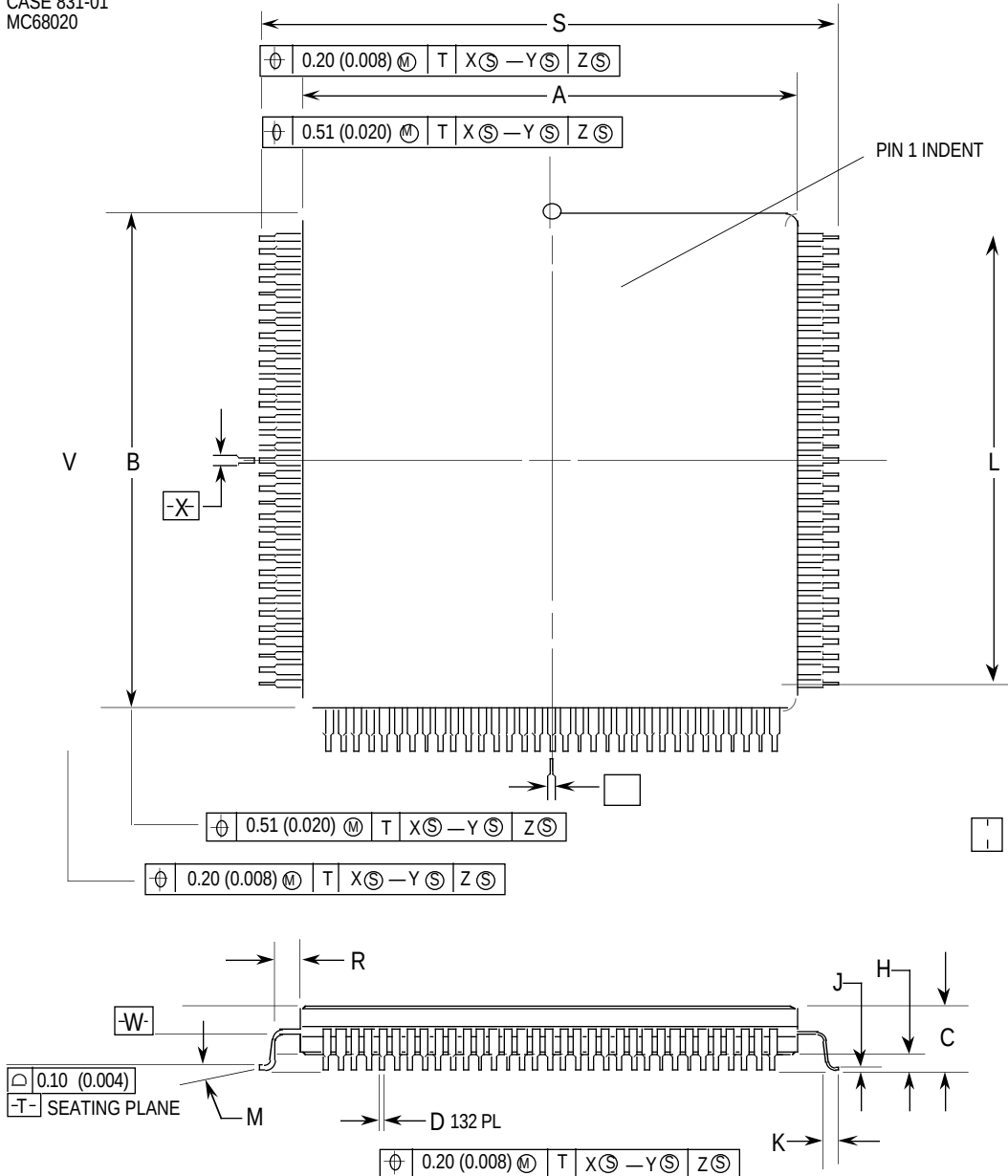


DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	24.06	24.20	0.947	0.953
B	24.06	24.20	0.947	0.953
C	4.07	4.57	0.160	0.180
D	0.21	0.30	0.008	0.012
G	0.64 BSC		0.025 BSC	
H	0.51	1.01	0.020	0.040
J	0.16	0.20	0.006	0.008

- NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
 2. CONTROLLING DIMENSION: INCH
 3. DIMENSIONS A, B, N, AND R DO NOT INCLUDE MOLD PROTRUSION ALLOWABLE MOLD PROTRUSION FOR DIMENSIONS A AND B IS 0.25 (0.010), FOR DIMENSIONS N AND R IS 0.18 (0.007).
 4. DATUM PLANE -W- IS LOCATED AT THE UNDERSIDE OF LEADS WHERE LEADS EXIT PACKAGE BODY.
 5. DATUMS -X-, -Y-, AND -Z- TO BE DETERMINED WHERE CENTER LE

11.2.6 MC68020 FE Suffix—Package Dimensions

FE SUFFIX
CASE 831-01
MC68020

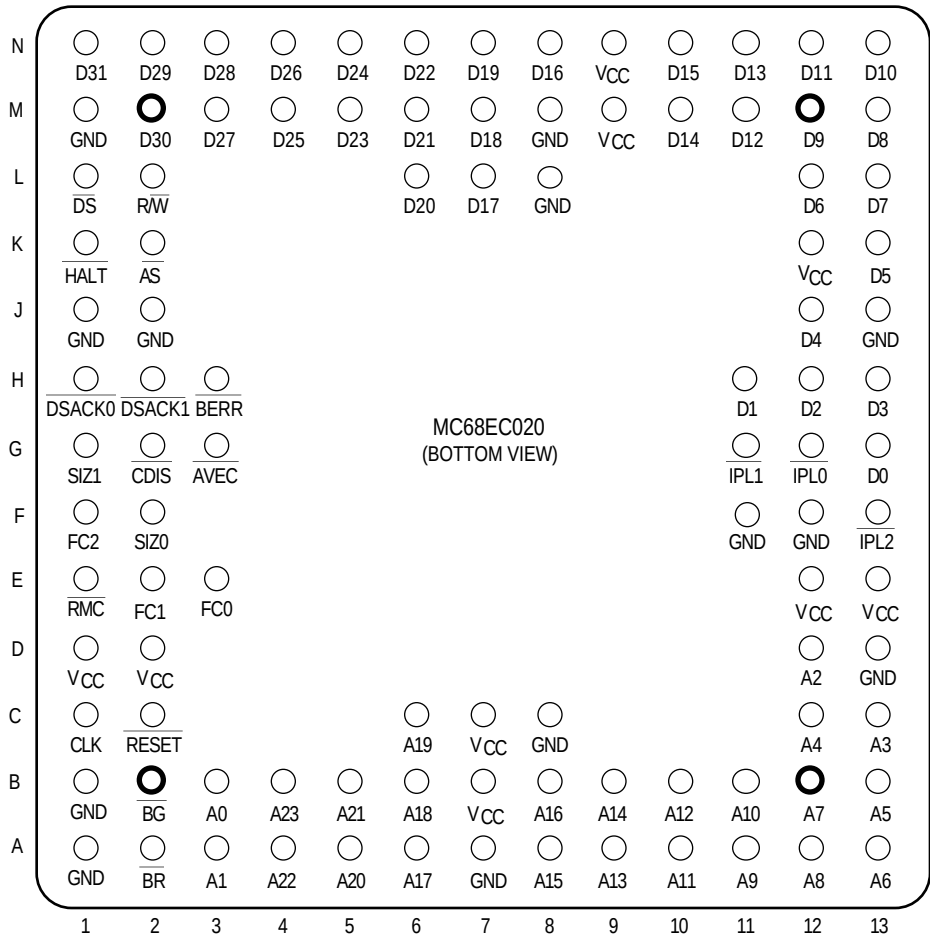


DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	21.85	22.86	0.860	0.900
B	21.85	22.86	0.860	0.900
C	3.94	4.31	0.155	0.170
D	0.204	0.292	0.0080	0.0115
G	0.64 BSC		0.025 BSC	
H	0.64	0.88	0.025	0.035
J	0.13	0.20	0.005	0.008
K	0.51	0.76	0.020	0.030
L	20.32 REF		0.800 REF	
M	0°	8°	0°	8°
R	0.64	—	0.025	—

NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH
3. DIMENSIONS A AND B DEFINE MAXIMUM CERAMIC BODY DIMENSIONS INCLUDING GLASS PROTRUSION AND MISMATCH OF CERAMIC BODY TOP AND BOTTOM.
4. DATUM PLANE -W- IS LOCATED AT THE UNDERSIDE OF LEADS WHERE LEADS EXIT PACKAGE BODY.
5. DATUMS -X-, -Y-, AND -Z- TO BE DETERMINED WHERE CENTER LEADS EXIT PACKAGE BODY AT DATUM -W-.
6. DIMENSIONS S AND V TO BE DETERMINED AT SEATING PLANE, DATUM -T-.
7. DIMENSIONS A AND B TO BE DETERMINED AT DATUM PLANE -W-.

11.2.7 MC68EC020 RP Suffix—Pin Assignment

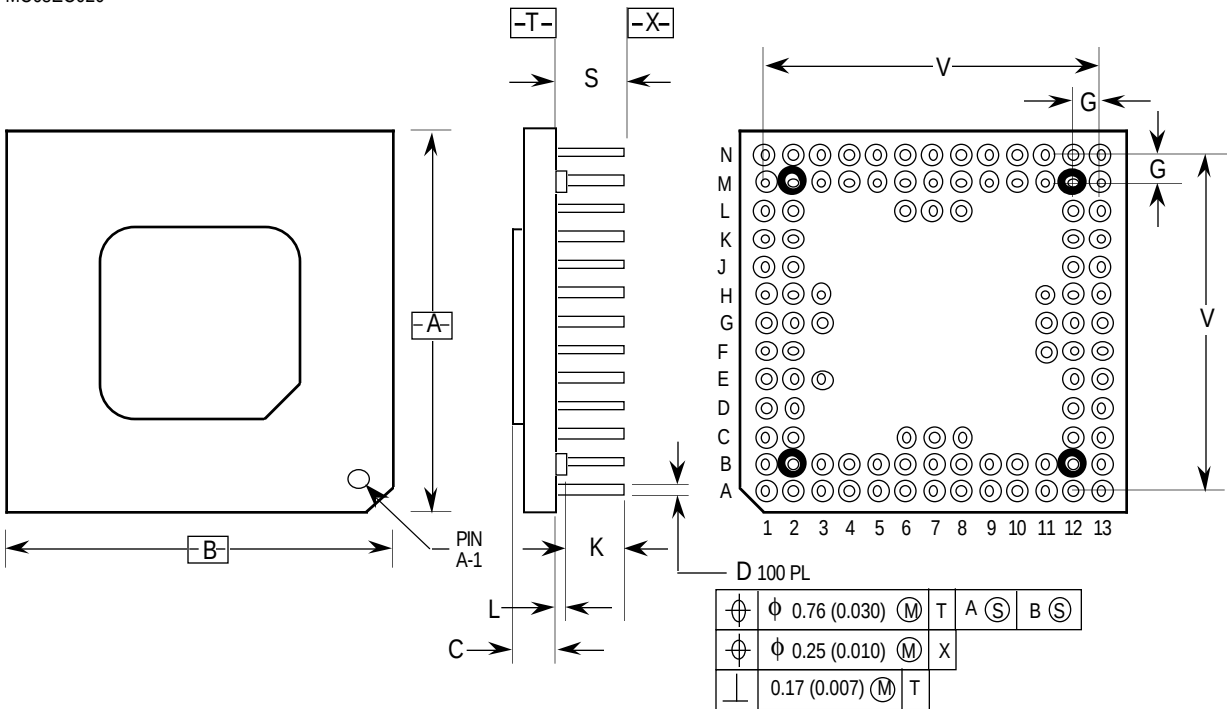


The V_{CC} and GND pins are separated into four groups to provide individual power supply connections for the address bus buffers, data bus buffers, and all other output buffers and internal logic. It is recommended that all pins be connected to power and ground as indicated.

Group	V_{CC}	GND
Address Bus	B7, C7	A1, A7, C8, D13
Data Bus	K12, M9, N9	J13, L8, M1, M8
Logic	D1, D2, E12, E13	F11, F12, J1, J2
Clock	—	B1

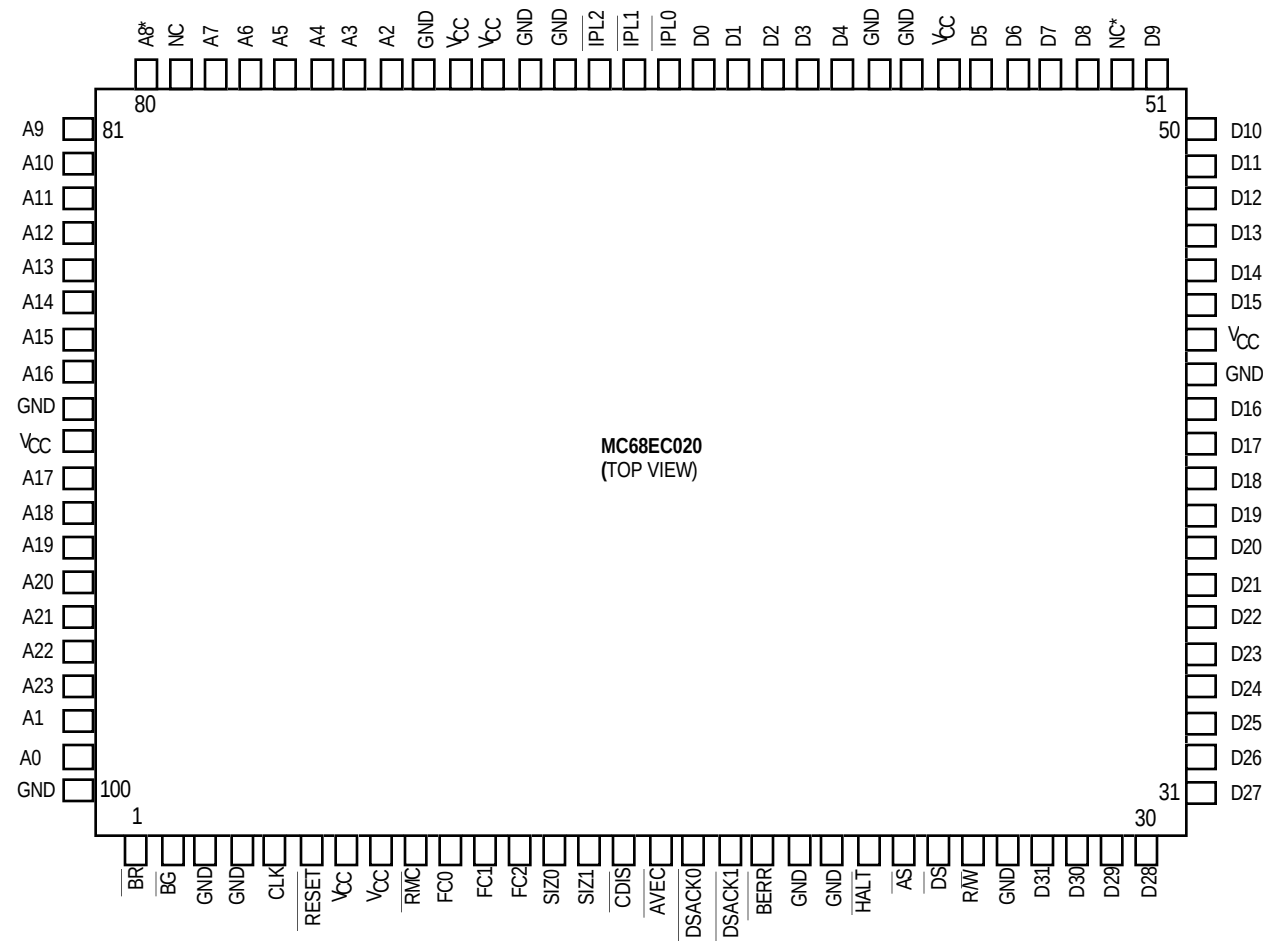
11.2.8 MC68EC020 RP Suffix—Package Dimensions

RP SUFFIX
CASE 789H-01
MC68EC020



- NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
 2. CONTROLLING DIMENSION: INCH
 3. DIMENSION D INCLUDES LEAD FINISH.

11.2.9 MC68EC020 FG Suffix—Pin Assignment



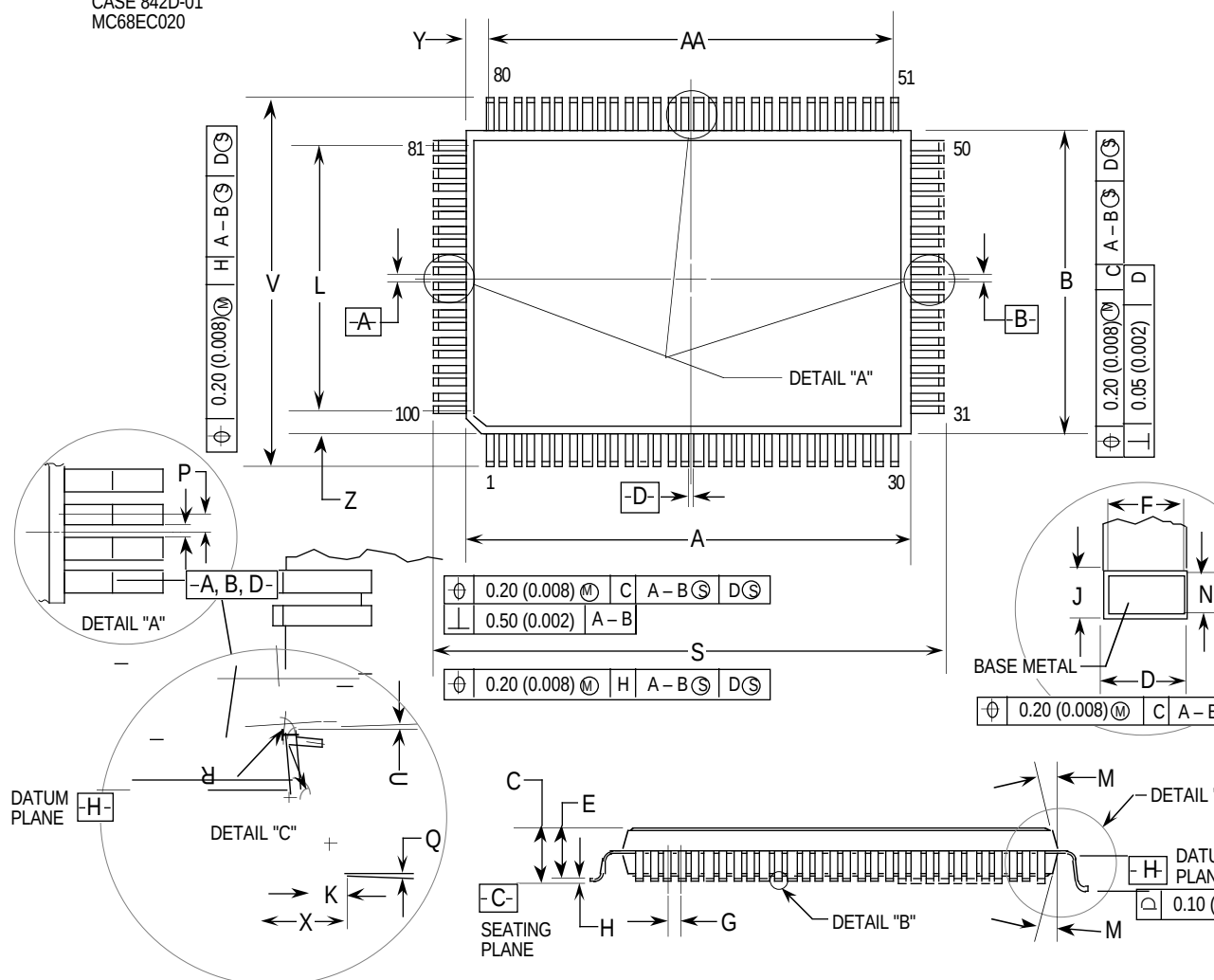
*NC—Do not connect to this pin.

The V_{CC} and GND pins are separated into four groups to provide individual power supply connections for the address bus buffers, data bus buffers, and all other output buffers and internal logic. It is recommended that all pins be connected to power and ground as indicated. NC pins are reserved by Motorola for future use and should have no external connection.

Group	V_{CC}	GND
Address Bus	90	72, 89, 100
Data Bus	44, 57	26, 43, 58, 59
Logic	7, 8, 70, 71	3, 20, 21, 68, 69
Clock	—	4

11.2.10 MC68EC020 FG Suffix—Package Dimensions

FG SUFFIX
CASE 842D-01
MC68EC020



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	19.90	20.10	0.783	0.791
B	13.90	14.10	0.547	0.555
C	—	3.30	—	0.130
D	0.22	0.38	0.009	0.015
E	2.55	3.05	0.100	0.120
F	0.22	0.33	0.009	0.013
G	0.65 BSC		0.026 BSC	
H	0.10	0.36	0.004	0.014
J	0.13	0.23	0.005	0.009
K	0.65	0.95	0.026	0.037
L	12.35 REF		0.486 REF	
M	5°	16°	5°	16°
N	0.13	0.17	0.005	0.007
P	0.325 BSC		0.013 BSC	
Q	0°	7°	0°	7°
R	0.25	0.35	0.010	0.014
S	23.65	24.15	0.931	0.951
T	0.13	—	0.005	—
U	0°	—	0°	—

NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETER.
3. DATUM PLANE **-H** IS LOCATED AT BOTTOM OF LEAD AND IS COINCIDENT WITH THE LEAD WHERE THE LEAD EXITS THE PLASTIC BODY AT THE BOTTOM OF THE PARTING LINE.
4. DATUMS **-A**, **-B**, AND **-D** TO BE DETERMINED AT DATUM PLANE **-H**.
5. DIMENSIONS S AND V TO BE DETERMINED AT SEATING PLANE **-C**.
6. DIMENSIONS A AND B DO NOT INCLUDE MOLD PROTRUSION. ALLOW PROTRUSION IS 0.25 (0.010) PER SIDE. DIMENSIONS A AND B DO INCLUDE MOLD MISMATCH AND ARE DETERMINED AT DATUM PLANE **-H**.
7. DIMENSION D DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.08 (0.003) TOTAL IN EXCESS OF THE DIMENSION AT MAXIMUM MATERIAL CONDITION. DAMBAR CANNOT BE LOCATED ON THE LOWER RADIUS OR THE FOOT